

# MSC ELSA 3 — floating debris & pellet drift

As of 27 May, 12:00 IST · Prepared by DriftSight · prototype

**Summary.** 1 satellite pass processed since the sinking on 25 May 2025 (1 usable, 0 clouded). DriftSight has flagged 3 candidate debris patches; none is confirmed yet (confirmation needs a second clear look), and no look-alikes have been rejected yet. The pellet forecast puts 1% of the simulated nurdles ashore by this time.

|                          |                      |                           |                     |
|--------------------------|----------------------|---------------------------|---------------------|
| <b>0</b>                 | <b>0</b>             | <b>1%</b>                 | <b>2</b>            |
| confirmed debris patches | look-alikes rejected | pellets ashore (forecast) | high-priority zones |

## TRACKED PATCHES

| Patch   | First seen        | Area                 | Confidence | Status |
|---------|-------------------|----------------------|------------|--------|
| Patch A | 27 May, 11:00 IST | 3,300 m <sup>2</sup> | 66%        | watch  |
| Patch B | 27 May, 11:00 IST | 300 m <sup>2</sup>   | 49%        | watch  |
| Patch C | 27 May, 11:00 IST | 3,200 m <sup>2</sup> | 66%        | watch  |

## PELLET LANDFALL OUTLOOK

Thiruvananthapuram 70% (first 27 May) · Kanyakumari 16% (first 29 May) · Kollam 2% (first 27 May) · Ramanathapuram 0% (first 7 Jun). Forecast first arrival at Kanyakumari 29 May, 04:00 IST, consistent with the 30 May beach survey that rated pellet pollution “Very High”.

## PRIORITY ACTIONS

- HIGH** — Kanyakumari coast (8.211°N 77.205°E): Beach survey and manual pellet collection; alert fishing villages. Nearest harbour Colachel.
- HIGH** — Thiruvananthapuram coast (8.494°N 76.899°E): Beach survey and manual pellet collection; alert fishing villages. Nearest harbour Vizhinjam.
- MEDIUM** — Patch A (at sea) (8.709°N 76.499°E): Boat verification, then boom / skimmer recovery if confirmed. Nearest harbour Neendakara.
- MEDIUM** — Patch C (at sea) (8.817°N 76.236°E): Boat verification, then boom / skimmer recovery if confirmed. Nearest harbour Neendakara.

## MISSIONS

No missions created yet.

## DATA & METHOD

Detection: per-pixel classifier on Sentinel-2 bands with FDI/NDVI. Tracking: ensemble particle drift with Bayesian confirmation at each pass. Prototype uses synthetic imagery and a synthetic monsoon circulation; production uses Sentinel-2/Landsat/Sentinel-1, INCOIS/CMEMS currents and ERA5 winds.

Prototype. Detection imagery and ocean fields in this report are simulated; no figure here is a measurement. Production sources: Sentinel-2 / Landsat / Sentinel-1 imagery, INCOIS and CMEMS currents, ERA5 winds.